A rotational obstruction to nonlinear continuation of Einstein–scalar shell perturbations

Ricardo Maldonado
Independent Researcher
ORCID: 0009-0009-3937-6527
ricardomaldonadoandrade@gmail.com

13 October 2026 (America/Los_Angeles) — technical draft for author review

Abstract

We derive a quadratic obstruction to nonlinear continuation of actual smooth linearized solutions in a five-dimensional Einstein–scalar model with two interiors and one dynamical timelike shell. An unrestricted boundary variation retains both outward Gibbons–Hawking–York terms and the common tension. Its completed Noether cut potential is explicitly proportional to the Israel residual, proving cancellation for common shell-tangent diffeomorphisms. Moving embeddings are handled by exact pullbacks. Differentiation at the background gives ten necessary Lorentz quadratic conditions for parameter-C^2 exact families with closed matched Cauchy sections. Independently, an ADM momentum-plus-Israel calculation retains arbitrary shell tilt and proves that no second-order shell correction can cancel the rotational constraint Hessian. Compact degree-two data yield a nonzero Hessian and hence a nonintegrable smooth non-gauge linear state, which can also undergo shell contact. Within an eighteen-dimensional quadrupole data space with fixed radial profile, time scale and reflection parity, all ten necessary tests vanish exactly when the displacement and velocity matrices commute. Passing these tests does not prove second-order solvability or nonlinear integrability.

Status. Technical continuation for author review, with companion proofs and executable algebra checks. This is not an APS submission or external peer review. The previous linear shell-contact capsule [1] is unchanged.

1. Model, hypotheses and result

The first-order construction [1] supplies smooth finite-angular solutions of the complete Einstein–scalar junction problem, including states that reach the shell. It does not establish that every such solution is tangent to an exact nonlinear family. We prove an explicit obstruction to that extension under a specified classical regularity assumption.

Use signature $(-++++)$, $8\pi G_5 = 1$, outward spacelike normals n_i, and $K_{i,ab} = h_a{}^A h_b{}^B \nabla_A n_{i,B}$. The action is

$$S = \sum_{i=\pm} \int_{M_i} \epsilon_i \left[\frac{R_i}{2} - \frac{(\nabla\phi_i)^2}{2} - U(\phi_i) \right] + \int_W \epsilon_W \left[K_+ + K_- - \sigma(\varphi) \right]. \tag{1}$$

There are two gravitational boundary completions and **one** shell tension. The common induced fields obey the exact configuration requirements

$$X_+^* g_+ = X_-^* g_- = h, \qquad X_+^* \phi_+ = X_-^* \phi_- = \varphi. \tag{2}$$

The fixed functions are inherited from [1]: with $z = \phi - 1$,

$$\begin{gathered} W = \tfrac{1}{3} + z^2 + \tfrac{1}{3} z^3, \qquad U = \tfrac{1}{2} W_{,\phi}^2 - \tfrac{2}{3} W^2, \\ \sigma = 2W + \mathfrak{d}(1 + c_* + c_* z), \quad c_* = \frac{2}{1.0357712571566784} - \frac{4}{3}, \quad 0 < \mathfrak{d} \le 0.002. \end{gathered} \tag{3}$$

The decimal is exact; fix one background on the registered regular branch, without a new branch certification or a uniform zero-detuning limit. It has

$$g_0 = \Omega(s)^2 \eta, \quad \phi_0 = \Phi(s), \quad s = R^2 - T^2, \quad f = \Phi_s \neq 0, \quad \Omega > 0, \qquad W: \ s = b^2. \tag{4}$$

All coefficients are the actual background functions on the controlled interval of [1]. Its shell hypotheses include $a_b = \sigma_b/6 > 0$, $p_b = -\sigma_b'/2 \neq 0$ and $B_b = p_b'/p_b + \sigma_b''/2 > 0$, where $a = \rho'/\rho$, $p = \phi'$ in the radial proper-distance chart.

The complete Cauchy section comprises two compact four-balls glued at their common closed S^3 cut. No exterior boundary or cone excision is present; the smooth cone and its vertex remain internal. A hypothetical exact family is assumed twice continuously differentiable in its parameter in a piecewise C^k, $k \geq 4$, norm on a compact reference slab, with smooth timelike embeddings admitting collars, uniformly nondegenerate metrics and uniformly spacelike Cauchy sections. Pullbacks preserve (2) throughout the family. All bulk equations and full metric/scalar junctions hold. These are sufficient classical assumptions for the products, traces and derivatives below; no solution manifold or weaker-space extension is presumed.

Under these hypotheses, common shell-tangent diffeomorphisms are null directions of the completed presymplectic form. The ten background Lorentz symmetries consequently impose necessary quadratic conditions. We exhibit a smooth, non-gauge linear solution violating a rotational condition, allowing arbitrary second-order corrections in every bulk and shell sector. Within the restricted quadrupole data space of Section 6, all ten conditions vanish if and only if two real symmetric matrices commute.

2. Unrestricted boundary variation and cut cancellation

Fix the bulk Lee–Wald potential of [1]. For $q_{AB} = \delta g_{AB}$, $q = g^{AB} q_{AB}$ and $q^{AB} = g^{AC} g^{BD} q_{CD}$,

$$\theta^A = \sqrt{-g}\left[\tfrac{1}{2}(\nabla_B q^{AB} - \nabla^A q) - \nabla^A \phi\, \delta\phi\right], \qquad \omega = \delta\theta. \tag{5}$$

Each outward boundary form is represented in a common co-oriented density convention on W, denoted ϵ_W. The conversion is applied to θ_i, Q_i, β_i as well as GHY. Using raw oppositely oriented pullbacks together with unsigned Israel sums would be inconsistent.

Put $p_i = n_i \phi_i$ and $v_i^a = h^{ab} n_i^A q_{i,Ab}$. The unrestricted GHY identity is

$$\theta_i|_W + \delta(K_i \epsilon_W) = -\epsilon_W\left[\tfrac{1}{2}(K_i^{ab} - K_i h^{ab})\delta h_{ab} + p_i \delta\varphi\right] + d_W \beta_i, \quad \beta_i = -\tfrac{1}{2} i_{v_i} \epsilon_W. \tag{6}$$

Normal and mixed metric variations have not been gauge-fixed away. Define

$$\begin{gathered} I_{ab} = \sum_i (K_{i,ab} - K_i h_{ab}) + \sigma h_{ab}, \qquad P = p_+ + p_- + \sigma', \\ \ell = (K_+ + K_- - \sigma)\epsilon_W, \quad \beta = \sum_i \beta_i, \quad \alpha = -\epsilon_W[\tfrac{1}{2} I^{ab}\delta h_{ab} + P\delta\varphi]. \end{gathered} \tag{7}$$

Both sides share the variations in (7) because of (2). Summing (6) and varying the single tension gives $\sum_i \theta_i|_W + \delta\ell = \alpha + d_W\beta$.

Fix reference bulk vectors ξ_i tangent to W and inducing the same vector ξ there. Their normal derivatives are arbitrary. Let $G_\xi \Phi_i = \mathcal{L}_{\xi_i}\Phi_i$ and $Q_{\xi,i} = -\frac{1}{2} *_i d\xi_i^\flat$. Minimal scalar coupling adds no separate Komar potential. The essential local cancellation is

$$B_\xi := \sum_i Q_{\xi,i}|_W + i_\xi \ell - \beta(G_\xi) = -i_{I\cdot\xi}\epsilon_W. \tag{8}$$

Indeed, if $A_i^a = h^a{}_B n_i^A \nabla_A \xi_i^B$, then $Q_{\xi,i}|_W = -\frac{1}{2} i_{A_i + K_i\cdot\xi}\epsilon_W$ and $\beta_i(G_\xi) = -\frac{1}{2} i_{A_i - K_i\cdot\xi}\epsilon_W$. Subtracting cancels the normal extension A_i; the remaining GHY and tension terms give precisely I. Thus (8) is a residual identity, not an assumption that the boundary charge vanishes.

For $k_\xi = \sum_i (\delta Q_{\xi,i} - i_{\xi_i}\theta_i)|_W$, covariance, fixed parameters and Cartan's identity yield

$$k_\xi - (\delta\beta)(\delta, G_\xi) = \delta B_\xi - i_\xi \alpha + d_W(i_\xi \beta(\delta)). \tag{9}$$

The final term integrates to zero on the closed cut S. The timelike boundary's future-end orientation is opposite to ∂C_i, giving

$$\Theta_C^{\mathrm{rel}} = \sum_i \int_{C_i} \theta_i - \int_S \beta, \qquad \Omega^{\mathrm{rel}} = \delta\Theta_C^{\mathrm{rel}}. \tag{10}$$

The corner is included once. Write $E_{i,AB} = G_{i,AB} - T_{i,AB}$, $C_{\xi,i} = E_i{}^A{}_B \xi_i^B \epsilon_{i,A}$ and $\boldsymbol{E}_i \delta\Phi_i = \epsilon_i[-E_i^{AB}\delta g_{i,AB}/2 + (\Box\phi_i - U')\delta\phi_i]$. The bulk identity and (9) give

$$\begin{aligned} H_\xi &= \sum_i \int_{C_i} C_{\xi,i} + \int_S B_\xi, \\ \Omega^{\mathrm{rel}}(\delta, G_\xi) &= \delta H_\xi + \sum_i \int_{C_i} i_{\xi_i}(\boldsymbol{E}_i \delta\Phi_i) - \int_S i_\xi \alpha. \end{aligned} \tag{11}$$

On a full solution and a full tangent solution, all residuals and their linearizations vanish. Hence $\Omega^{\mathrm{rel}}(\delta, G_\xi) = 0$. Curling the boundary variation similarly gives conservation of (10).

Embedding consistency is also nonlinear. Gauss–Codazzi and common induced fields imply, with $\Delta K_{ab} = K_{+,ab} - K_{-,ab}$ and $\Delta p = p_+ - p_-$,

$$\begin{aligned} \sum_i E_{i,n_i a} &= D_b I^b{}_a - P D_a \varphi, \\ \sigma \Delta K + \sigma' \Delta p &= 2(E_{+,n_+n_+} - E_{-,n_-n_-}) + I_{ab}\Delta K^{ab} + P\Delta p. \end{aligned} \tag{12}$$

Thus tangential conservation and normal force are retained residual relations, rather than extra equations dropped using a first-order identity.

3. Moving embeddings and the ten necessary conditions

Straighten each moving region by $Y_i : \bar{M}_i \to M_i$, with $Y_i|_W = X_i$, and set $\chi_i = \delta Y_i \circ Y_i^{-1}$. The exact action equals its barred version, and

$$\delta\bar{\Phi}_i = Y_i^*(\delta\Phi_i + \mathcal{L}_{\chi_i}\Phi_i), \quad \Theta_C^{\mathrm{rel}} = \sum_i \int_{Y_i(\bar{C}_i)} (\theta_i + i_{\chi_i} L_i + C_{\chi_i,i}) + \int_S \sum_i Q_{\chi_i,i} - \int_{\bar{S}} \bar{\beta}. \tag{13}$$

Pullbacks and the common orientation conversion are understood in the cut terms. The $i_\chi L$ term records moving domains; it generally survives with matter. The Q_χ and $\bar{\beta}$ terms retain the gravitational cut. C_χ is removed only on shell, and from the curl only with the linearized constraints. Definition (10) in barred variables fixes that curl without omitting moving-location or normal-normalization variations. Twice differentiating (2) includes bending and evaluation terms: for example, $\partial_\varepsilon^2(Y_\varepsilon^* g_\varepsilon) = Y_\varepsilon^*(\ddot{g} + 2\mathcal{L}_\chi \dot{g} + \mathcal{L}_{\dot{\chi}} g + \mathcal{L}_\chi^2 g)$.

Since $f \neq 0$, a common metric–scalar Killing field satisfies $Ks = 0$ and then $\mathcal{L}_K \eta = 0$. Flat translations fail $Ks = 0$ on an open set; the remaining algebra is exactly the six rotations and four boosts of $\mathfrak{so}(1,4)$. The shell requires the same action on both copies. This counts background symmetries, not every possible adjoint kernel in an unspecified constraint domain.

Use the fixed compensator $k_0 = K|_W$:

$$G_{K,k_0} X_i = -K \circ X_i + (X_i)_* k_0, \qquad G_{K,k_0}(\Phi_0) = 0. \tag{14}$$

Extend k_0 by fixed shell-tangent reference vectors ξ_i, equal to the background K when $Y_{i0} = \mathrm{id}$. The compensated transformation then gives $G_{K,k_0}\bar{\Phi}_i = \mathcal{L}_{\xi_i}\bar{\Phi}_i$, so (11) applies to the moving system through its barred action. Its linear action $\mathcal{K} = DG_{K,k_0}|_0$ is the Lie derivative on the reconstructed fields and their normal graphs. For an exact family $\Phi_\varepsilon = \Phi_0 + \varepsilon v + \varepsilon^2 w/2 + o(\varepsilon^2)$, differentiate the proved degeneracy along that family:

$$\Omega_0^{\mathrm{rel}}(v, \mathcal{K}v) = 0, \qquad D^2 H_K|_0[v,v] = \Omega_0^{\mathrm{rel}}(v, \mathcal{K}v). \tag{15}$$

The Hessian uses a chart on the exact matching space. Equation (11) gives $DH_K|_0[w] = 0$ for arbitrary matching variations; derivatives of its residual terms vanish for v. Thus (15) also follows by differentiating the exact constraint. A Taub functional defined as the Taylor coefficient is half this Hessian.

Our relative convention is canonical: $\Omega_{\mathrm{future}} = -\Omega^{\mathrm{rel}}$. For the compact initial-support states below, choose the first variation of the straightening map zero near the **entire initial Cauchy section**, including its required time jet. All extra terms in (13) then drop from the initial quadratic pairing, and $\Omega_{\mathrm{future}} = \Omega_{\mathrm{phys}}$ of [1]. Vanishing merely at the cut would not remove bulk $i_\chi L$ terms. No new component-by-component identification of arbitrary after-contact cut currents is claimed; the established total-current conservation of [1] already transports its initial pairing through contact.

4. Independent ADM derivation, with arbitrary shell tilt

The rotational obstruction admits a separate constraint proof. Let γ be the spatial metric, u its future unit normal and k its extrinsic curvature. The normalization in (1) gives

$$\pi^{ab} = \frac{\sqrt{\gamma}}{2}(k^{ab} - k\gamma^{ab}), \quad \varpi = \sqrt{\gamma}\, u\phi, \quad \mathcal{H}_a = -2D_b\pi^b{}_a + \varpi D_a\phi = -\sqrt{\gamma}\, E_{ua}. \tag{16}$$

Let s_i be the outward spatial normal to the cut, τ the common future normal to it within the shell, and c its common metric. Define the boost angle η_i by

$$n_i = \cosh\eta_i\, s_i + \sinh\eta_i\, u_i, \qquad \tau = \cosh\eta_i\, u_i + \sinh\eta_i\, s_i. \tag{17}$$

For a fixed common cut-tangent rotation J, differentiating this normal frame along J gives $k_i(s_i, J) = -K_i(\tau, J) - J\eta_i$. Also $I(\tau, J) = \sum_i K_i(\tau, J)$: the trace and pure-tension terms have zero mixed projection. Integration by parts therefore derives the exact functional

$$\begin{aligned} \mathcal{C}_J &:= \sum_i \int_{C_i} \mathcal{H}_{i,a} J^a - \int_S \sqrt{c}\, I(\tau, J) \\ &= M_J + \int_S \sqrt{c}\, J(\eta_+ + \eta_-), \qquad M_J = \sum_i \int_{C_i} (\pi_i^{ab}\mathcal{L}_J\gamma_{i,ab} + \varpi_i J\phi_i). \end{aligned} \tag{18}$$

Every exact constraint-and-junction datum has $\mathcal{C}_J = 0$. The angle term is essential for a tilted cut. Scalar-dependent tension has no independent intrinsic kinetic momentum or extra mixed angular stress.

At $T = 0$, $\pi_0 = \varpi_0 = \eta_{i0} = 0$ and $\mathcal{L}_J\gamma_0 = J\phi_0 = J\sqrt{c_0} = 0$. For any second-order correction w, even one nonzero on the cut,

$$D\mathcal{C}_J|_0[w] = \int_S \sqrt{c_0}\, J\eta_{\rm sum}[w] = 0. \tag{19}$$

The equality uses the closed cut and its invariant background density. For a first-order state vanishing on an open shell neighborhood, the corner Hessian $2\int_S \delta\sqrt{c}\, J\delta\eta_{\rm sum}$ is zero. Thus

$$0 = D^2\mathcal{C}_J[v, v] = 2\sum_i \int_{C_i} [\delta\pi_i^{ab}\mathcal{L}_J\delta\gamma_{i,ab} + \delta\varpi_i J\delta\phi_i]. \tag{20}$$

The support restriction removes this *quadratic* cut term; it does not remove w. Equation (19) is the separate cancellation of its arbitrary second-order shell contribution. No scalar-sector restriction is imposed on w.

For master initial data $q = F|_0$, $p = F_T|_0$, the inherited reconstruction is

$$\delta g_{AB} = 2F(s\eta_{AB} - 3X_AX_B), \qquad \delta\phi = -\frac{3(E+4)F}{2\Omega^2 f}, \quad E = X^A\partial_A. \tag{21}$$

Direct ADM variation at $T = 0$, with $D = r\partial_r$, gives

$$\begin{aligned} \delta\gamma_{ij} &= 2q(r^2\delta_{ij} - 3x_ix_j), \qquad \delta\pi^{ij} = -\frac{3p}{2\Omega}x_ix_j, \\ \delta\phi &= -\frac{3(D+4)q}{2\Omega^2 f}, \qquad \delta\varpi = -\frac{3\Omega(D+5)p}{2f}. \end{aligned} \tag{22}$$

The $D + 5$ follows from $[\partial_T, E] = \partial_T$. Substitution in (20) recovers the actual Cauchy bilinear

$$\mathcal{B}(q, p) = \int dr\, d\Omega_3 [m_r(D+4)q(D+5)p + n_rqp], \quad m_r = \frac{9r^3}{4\Omega f^2}, \quad n_r = \frac{6r^7}{\Omega}. \tag{23}$$

Per copy, [1] has $\Omega_1(v_1, v_2) = \mathcal{B}(q_1, p_2) - \mathcal{B}(q_2, p_1)$. The canonical $\delta\pi \wedge \delta\gamma + \delta\varpi \wedge \delta\phi$ orientation is its negative. For two copies of fixed reflection parity,

$$D^2\mathcal{C}_J[v, v] = 4\mathcal{B}(Jq, p) = -\Omega_{\rm phys}(v, Jv). \tag{24}$$

This independently fixes both the sign and the Hessian factor two.

5. A smooth non-gauge state with nonzero obstruction

Choose a real harmonic H with $JH \neq 0$, a nonzero real $Q \in C_c^\infty(I)$, and a fixed real $\omega \neq 0$ with inverse-time units. Take

$$q = QH, \qquad p = \omega QJH. \tag{25}$$

Angular integration by parts gives, without assuming symmetry of the radial bilinear,

$$D^2\mathcal{C}_J[v,v] = 4\omega\mathcal{B}_{\rm rad}(Q,Q)\int_{S^3}(JH)^2. \tag{26}$$

Here $\mathcal{B}_{\rm rad}$ is (23) without angular integration. With $L = \Omega_s/\Omega + 2f_s/f$, compact radial integration gives

$$\mathcal{B}_{\rm rad}(Q,Q) = \int_I [m_r\{(DQ)^2 + (2+9r^2L)Q^2\} + n_rQ^2]\,dr \geq \left(\frac{\pi^2 m_*}{w_I^2} - M_*\right)\int_I Q^2 dr > 0, \tag{27}$$

where $m_* = \min_{\bar{I}}(m_r r^2) > 0$, $M_* = \max_{\bar{I}} m_r(-2-9r^2L)_+$ and w_I is the annulus width. Choose I sufficiently thin within the admissible collar of [1]. This is positivity of one radial diagonal, not a positive gravitational energy statement.

An explicit quadrupole uses $n = x/r$ and

$$\begin{gathered} H = n_1^2 - n_2^2, \quad J = x_1\partial_2 - x_2\partial_1, \quad JH = -4n_1n_2, \quad J^2H = -4H, \\ \int H^2 = \pi^2/3, \quad \int (JH)^2 = 4\pi^2/3, \qquad \boxed{D^2\mathcal{C}_J[v,v] = \frac{16\pi^2}{3}\omega\mathcal{B}_{\rm rad}(Q,Q) \neq 0.} \end{gathered} \tag{28}$$

Either reflection parity doubles the same quadratic contribution. Equations (19) and (28) exclude a parameter-C^2 exact coupled family with this first derivative, including families with unrestricted second-order sectors.

These are actual solutions of the inherited finite-angular linear problem, not formal unconstrained tensors. The compact support satisfies every initial shell compatibility condition; [1] proves smooth evolution and full junction sufficiency. In its hemisphere collar coordinate x, choose support $[a_0, d_0]$ and a slab with $0 < a_0 < d_0 < \tau_*$, $d_0 + \tau_* < \pi/2$, and coefficients within the controlled background interval. The support infimum is a_0. The displacement harmonic H and velocity harmonic JH are orthogonal since $\int HJH = 0$. Their separate boundary activations proved in [1] therefore cannot cancel as a full angular trace. The obstructed state can be chosen among smooth contacting solutions; no nonzero jet at the exact support front is asserted.

For $l = 2$, non-gauge character follows from the angular trace-free metric and mixed components in (21). A scalar gauge covector with angular part ZD_aH_l would produce $2Z(\mathrm{Hess}\,H_l)^{\rm TF}$, whose squared norm is $\frac{2}{3}\lambda_l(\lambda_l - 3)\int H_l^2 > 0$. The absent trace-free part forces $Z = 0$, and the absent mixed part then forces the remaining scalar gauge coefficients to zero. Other angular sectors cannot cancel this projection.

6. Quadrupole classification of the ten tests

Fix the same Q, ω and reflection parity. Every real degree-two harmonic is $H_A(n) = n^TAn$ for a symmetric traceless 4×4 matrix. Consider the eighteen-dimensional data space

$$q = QH_A, \qquad p = \omega QH_B, \quad A = A^T,\ B = B^T,\ \mathrm{tr}\,A = \mathrm{tr}\,B = 0. \tag{29}$$

For antisymmetric M, put $J_M = (Mn)\cdot\partial_n$ and $V_3 = 2\pi^2$. Exact sphere moments give

$$\int H_AH_B = \frac{V_3}{12}\,\mathrm{tr}(AB), \qquad J_MH_A = H_{[A,M]}, \qquad D^2\mathcal{C}_{J_M}[v,v] = \frac{V_3}{3}\omega\mathcal{B}_{\rm rad}(Q,Q)\,\mathrm{tr}(B[A,M]). \tag{30}$$

For $J_{ij} = X_i\partial_j - X_j\partial_i$, this is $-(2V_3/3)\omega\mathcal{B}_{\rm rad}(Q,Q)[A,B]_{ij}$. Since $[A,B]$ is antisymmetric and the prefactor is nonzero, all six rotational tests vanish exactly when $[A,B] = 0$.

The four boosts vanish for every pair (29). The master equation gives $F_{TT}|_0 = \mathcal{A}q$, where $\mathcal{A} = \Delta_{\mathbb{R}^4} - 2LD - [8L + (8/3)r^2f^2]$. A boost acts on initial data by

$$(q,p) \longmapsto (X_ip,\ \partial_iq + X_i\mathcal{A}q). \tag{31}$$

Both entries in (29) have even antipodal parity; (31) has odd parity. The radial operators and weights in (23) preserve it, so each full-sphere pairing is zero. The transformed state lies in the enlarged finite-angular family; (29) itself is not asserted to be Lorentz invariant.

Thus **all ten necessary Lorentz tests vanish within (29) if and only if** $[A, B] = 0$. Commuting real symmetric matrices are simultaneously orthogonally diagonalizable. The degree-two specialization of the previous same-real-harmonic span is $A = aC$, $B = bC$, including pure displacement and pure velocity. It passes these tests, with nonlinear extension still unproved. Neither this commuting criterion nor ten zero charges characterizes the full nonlinear constraint range or guarantees higher-order compatibility.

7. Prior work, verification and declarations

The general Noether and linearization-stability mechanisms are established. Iyer–Wald [2] supplies the bulk Hamiltonian identity; Harlow–Wu [3] derives the completed boundary potential, but its GR example fixes an external induced metric. The physical interface here instead varies common induced fields and cancels the two sides against the single tension. Speranza [4] provides exact embedding pullback and moving-cut formulas; auxiliary edge-mode degeneracy alone is not used to prove (8)–(11).

Hájíček–Kijowski [5] is direct physical-shell precedent for induced matching, the area/boost-angle cut pair and interface constraints. Its erratum is essential: items 13–14 restore density factors and place the last integral of its Eqs.(99)–(100) on the external boundary, rather than the internal shell. Its fluid system and differentiability discussion are not silently imported as an Einstein–scalar existence theorem. Altas–Tekin [6] explicitly exhibits the smooth compact-Cauchy Taub obstruction. Classical Fischer–Marsden, Moncrief and Arms references, distinguished from originals actually read here, appear in the companion literature register. The recent field-dependent-generator analysis [7] reinforces the need for additional parameter variations if the compensator in (14) were field dependent; we keep it fixed. The contribution here is the explicit application to this matched action, its independent rotational normalization, and the evaluated finite-angular obstruction, without a claim of general or worldwide novelty.

The recorded normal and optimized-Python runs each contain **81 positive algebra checks and 21 wrong-formula controls**: 6/8 for nonlinear residuals and cut coefficients, 46/9 for the independent rotational calculation, and 29/4 for quadrupole algebra. The replay records identify both execution modes and their results. These are finite algebra checks, not machine-certified PDE, functional-analytic or complete differential-form proofs. The linear existence theorem is inherited from [1]; exact-family differentiability is an explicit hypothesis. No dynamical growth, positive physical energy, stability or cosmological-origin inference follows from the obstruction.

Code and data availability. Companion material is provided at relative capsule paths: nonlinear identity, independent ADM proof, quadrupole proof, and primary-source comparison. The original checkers are Noether algebra, rotational constraints, and quadrupoles. All paths are relative to the `HDBLAST_SHELL_NOETHER_20261013` capsule, continuation `2026.10.13-shell-noether`; its integrity and replay records identify the packaged inputs and executed checks.

Author and support. Ricardo Maldonado, Independent Researcher, ORCID 0009-0009-3937-6527; correspondence: ricardomaldonadoandrade@gmail.com. No external funding is reported. The author reports no known conflicts of interest.

AI assistance and review. Substantial OpenAI Codex assistance was used for derivation, code, literature work, internal review and drafting. Separate agent processes supplied internal checks; these are not external human peer review. The author retains responsibility for verification, interpretation and any submission. No APS submission, external peer review or novelty certification is claimed.

References

1. R. Maldonado, *Smooth Einstein–scalar perturbations through timelike shell contact*, technical checkpoint, 13 October 2026, frozen `HDBLAST_REGULAR_SHELL_CONTACT_20261013` capsule, especially the existence, full-junction and physical-current proofs.
2. V. Iyer and R. M. Wald, *Some Properties of Noether Charge and a Proposal for Dynamical Black Hole Entropy*, Phys. Rev. D **50**, 846–864 (1994), DOI, arXiv:gr-qc/9403028v1.

3. D. Harlow and J.-q. Wu, *Covariant phase space with boundaries*, JHEP **10** (2020) 146, DOI, arXiv:1906.08616v4.
4. A. J. Speranza, *Local phase space and edge modes for diffeomorphism-invariant theories*, JHEP **02** (2018) 021, DOI, arXiv:1706.05061v2.
5. P. Hájíček and J. Kijowski, *Lagrangian and Hamiltonian formalism for discontinuous fluid and gravitational field*, Phys. Rev. D **57**, 914–935 (1998), DOI, arXiv:gr-qc/9707020v1; erratum, Phys. Rev. D **61**, 129901 (2000), DOI.
6. E. Altas and B. Tekin, *Second Order Perturbation Theory in General Relativity: Taub Charges as Integral Constraints*, Phys. Rev. D **99**, 104078 (2019), DOI, arXiv:1903.11982v2.
7. H.-F. Ding, *Conserved charges and the first law of black holes for field dependent symmetry generators in the covariant phase space formalism* (2026), arXiv:2609.27454v2.